



Linking cognitive biases: The successes of a test case that predicted variations in endowment effect magnitudes

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Abstract

Cognitive biases are patterned decisions that deviate from the predictions of rational-choice models. They often have profound and adverse impacts on behavior, making them important to understand. The common approach to them catalogs empirical regularities that exist but lacks a sound theoretical foundation to explain and predict why the biases exist, why they are patterned as they are, and how we can best predict new and deeper patterns. The approach described here can provide such a foundation, link cognitive biases together (as a function of common underlying psychological processes), and increase our predictive power. This article reviews a 20-year arc of research that tested whether looking at cognitive biases through the lens of evolutionary biology—essentially, at how all brains are shaped to have behavioral leanings—can illuminate the origins of biases and thereby enable useful, testable predictions. The test case centers on the “endowment effect” and led to four interconnected discoveries about how, when, and where the endowment effect arises. It culminated in the successful prediction of a very large percentage of the variance in endowment effect magnitudes across a large new set of items—something never before accomplished. The key fact is that natural selection can bias decision-making toward choices that were rational in ancestral conditions but are mismatched to modern environments, yielding outcomes that are irrational yet predictably patterned. More broadly, the success of this test case suggests that many other cognitive biases may be similarly approached and potentially linked by this predictively useful framework.

Keywords Cognitive bias · Endowment effect · Evolution · Adaptation · Mismatch

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Much remains to be learned about the endowment effect.

Daniel Kahneman [in *Thinking Fast and Slow*, 2013, p. 299].

1 Introduction: Cognitive biases

In pursuit of their own self-interests, people often make choices that maximize utility, in light of stable preferences. But not always. Sometimes people behave irrationally. And sometimes lots of people behave irrationally in the same way, rather than in random ways. When they do, that can have profound negative effects on enormous swaths of human society.

Along with his key colleagues, this symposium issue's honoree, Nobelist Daniel Kahneman, drew significant attention to a wide variety of discrete patterns in how people behave irrationally. These patterns arise, it is now commonly thought, from various *cognitive biases*. That is, from decision-making leanings—in other words, behavior-biasing inclinations—in human psychology. Later researchers, many in a field that came to be known as behavioral economics, extended and deepened this important work on cognitive biases.

Generally speaking, different biases were discovered through a combination of intuition, curiosity, ambition, accident, and effort (Thaler, 2015). There is much to recommend this approach, and it does a lot of work. Still, it is generally considered advisable to construct scientific theories for the causes of patterned phenomena. And that's largely because—absent a theory that can generate specific predictions and thereby shine light on specific things to investigate—discovering new stuff inevitably involves some groping around in the dark.

In this article, I'll summarize a 20-year arc of research that sought to test whether the development of such a scientific theory could pack a demonstrably positive payoff, when partnered with behavioral economics. The test case involved the so-called *endowment effect*. And it turned out to be more effective and successful than my colleagues and I could reasonably have hoped at the outset. Building up from the foundation of evolutionary biology, we discovered specific things that no other extant theory of the endowment effect predicted. This demonstrates the concrete, practical usefulness of the approach, both for the endowment effect itself as well as for, potentially, other cognitive biases, many of which may have arisen from the same underlying, behavior-biasing evolutionary process.

2 Test case: The endowment effect

The endowment effect is the widespread tendency people have to value something they own—even something they have just that instant come to own—at a much higher price than the *maximum* amount they would have paid to buy it. This phenomenon can be found even before a person has had any time to develop, say, sentimental

attachment to, or experience with, an item. And it can inhere before a person has acquired one whit of new knowledge about what they now own. That's inconsistent with the rationality assumption in neoclassical economics. For you should not logically refuse to sell for \$10 something you just bought and would never have paid more than \$5 to buy.

That people nonetheless often behave illogically in that context has inspired hundreds of published endowment effect experiments. Over 1,000 academic publications express concern that the endowment effect is deeply problematic for governmental policies (Korobkin, 2014), largely because it impedes the efficient flow and distribution of goods, services, and marketable legal rights to everyone downstream (Coase, 1960; Zamir, 2015; Viscusi & Huber, 2012). It also undermines efforts to accurately compensate losses and the like (Knetsch & Sinden, 1984).

To date, the vast bulk of the empirical literature on the endowment effect typically reports, for different tradable goods and rights, two things. One is the prevalence of the effect, that is, the percentage of the test subjects who exhibited it. The other is the magnitude of the effect. Here, "magnitude" means the ratio of the *minimum* price at which one would sell and the *maximum* price at which one would buy. The larger the ratio, the larger the magnitude of the effect. For instance, if subjects on average wouldn't buy an item for more than \$5 and wouldn't sell it for anything less than \$10, that's a smaller magnitude of the endowment effect (a multiple of 2) than when subjects wouldn't buy for more than \$100, and wouldn't sell for less than \$2,000 (a multiple of 20).

One of the most consistent findings in the many hundreds of endowment effect studies is that the magnitudes of the effect vary considerably with the items that subjects are considering (Horowitz & McConnell, 2002; Sayman & Öncüler, 2005; Plott & Zeiler, 2007; Tunçel & Hammitt, 2014; Zeiler, 2018). For instance, experiments have shown that sellers of a hunting permit demanded nearly 7 times as much money to part with one as the maximum that buyers were willing to pay for one (Heberlein & Bishop, 1986). Yet sellers of a book demanded only 1.04 times as much as the maximum buyers would pay (Chapman & Johnson, 1995).

Importantly, it has been a mystery why the magnitudes of endowment effects vary at all and, more specifically, why they vary as they do, across the various goods and rights that have been investigated. To my knowledge, exactly 0 studies prior to the last of our experiments discussed below have successfully predicted any of the enormous variation in the magnitudes of endowment effects across a large and novel set of items. That may be, in part, because behavioral economists tend to be more interested in the patterns that exist than in why they exist. And it may also be, in part, because it is not (yet) traditionally part of behavioral economics to ground the study of cognitive biases in a scientifically sound and prediction-generating theory of where the psychological predisposition that results in the endowment effect comes from in the first place, and why that psychological predisposition plays out differently—from small magnitudes to large—in different kinds of items.

Which is not to say that there aren't some theories (Morewedge & Giblin, 2015). For instance, behavioral economists have long attempted to explain the endowment effect's deviation from rational choice predictions by claiming that it's caused by human tendencies to *loss aversion* (Kahneman et al., 1990, 1991, 2008). Which is

to say (as it is often said): losses loom larger than gains, in people's minds. Unfortunately, one has to admit that that explanation doesn't really get us very far. For one thing, it just renames the phenomenon one is trying to explain with another phenomenon that is itself unexplained. If the endowment effect is caused by loss aversion, then what causes loss aversion? Was it just coin-flipping odds whether humans would be loss averse and exhibit an endowment effect, or be gain averse and exhibit an anti-endowment effect?¹

When pressed to explain where loss aversion comes from, many point to *prospect theory*, as developed by Kahneman and Tversky in 1979 (Kahneman & Tversky, 1979; Köszegi & Rabin, 2006; Tversky & Kahneman, 1991). Yet prospect theory, useful as it is conceptually, mainly formalizes the observation that people respond differently to equivalent choices, depending on their initial reference points. In this context, the reference point is meaningfully different, so the theory goes, between when you own a good or right and when you don't own it. But that doesn't explain *why* ownership of a good creates a significant reference point compared to, say, the color, size, or any other feature of the good. Prospect theory consequently organizes, cumulates, and graphs the already-known reality of ownership's significance for reference-dependence without even hinting at the reason for its unique significance. So prospect theory specifically, like loss aversion generally, cannot provide a secure theoretical foundation for understanding and explaining *why* people respond differently to equivalent choices (Klass & Zeiler, 2013; McCaffery, 1994; Jones, 2018). I have argued that believing that the endowment effect is caused by loss aversion as a function of prospect theory is, put frankly, like believing that sexual behavior is caused by abstinence aversion as a function of lust theory (Jones, 2018).

I think this is more speedbump than barrier, however. For there is a way to shore up the theoretical foundation and continue exploring this frontier landscape that Kahneman and colleagues so valuably opened. Here, I'll summarize experimental results consistent with 4 novel predictions, which show a path toward better explaining, and therefore better predicting, endowment effect patterns. This will also illustrate an approach that might yield similar successes with other cognitive biases. The test case was undertaken for that express purpose.

3 Four successful endowment effect predictions

To galvanize reader interest in reading on, I will cut to the chase and quickly list four increasingly powerful payoffs from the four predictions this approach generated. In subsequent sections, I'll summarize the theoretical perspective and provide further details on the logic underlying our predictions, the experimental designs, the results, and the significance.

¹ For similar reasons, *substitutability* between goods, which is sometimes advanced as an explanation for the persistence and magnitude of endowment effects (Hanemann, 1991; Shogren et al., 1994), cannot constitute a scientific theory for the phenomena, inasmuch as identifying variables that are relevant (compare: plants grow better with sun and rain) does not alone constitute a causal explanation for *why* they are relevant. Especially since the substitutability of the good in question is presumably just as well known to subjects in the seller position as in the buyer position.

We began by studying chimpanzees. Why? Because if the endowment effect is not just a quirk of human psychology, as we suspected, then finding and studying the effect in a non-human species can increase both the generality and the strength of the theory, in the same way that the theory of gravity is stronger when it predicts not only the motion of objects on Earth, but also the motion of stars and planets in space. That is, a satisfactory theory of the endowment effect should be able to explain the phenomenon wherever it exists, not just in the human bubble.

Here are our predictions and results from experiments with a shifting set of colleagues, but always with my principal colleague in this, the eminent primatologist Sarah Brosnan:

Prediction: The endowment effect will be present in some other species, particularly close primate relatives. Result: We were the first to discover the endowment effect in any non-human species,² here chimpanzees. (The results were later replicated with orangutan and gorilla subjects.)

Prediction: Chimpanzees will exhibit the endowment effect more frequently when trading foods for foods, than when trading toys for toys. Result: Subjects were 14 times more likely to exhibit an endowment effect in the former condition.

Prediction: In chimpanzees, the endowment effect *for the very same item* will be more prevalent when that item can be used in context than when it can't. Result: We were able to turn the endowment effect on, and then fully off, by simply manipulating an object's contextual usefulness.

Prediction: In humans, the evolutionary salience score of an object will predict some of the variance in the magnitudes of endowment effects across a large set of novel items. Result: The evolutionary salience of an item predicted fully 52% of the variance in magnitudes across a large set of novel items.

It is important to note that no other extant theory of the endowment effect predicted any of these results, either individually or as a connected set. The theoretical foundation that led to these predictions and discoveries has several parts, to which we turn next.

4 Theoretical foundation part I: Biology, causes, and cognitive biases

To understand the theoretical foundation that led to these predictions about the endowment effect specifically, we first need to frame the underlying theory that connects evolutionary processes to cognitive biases generally. To do that, this section

² An experiment on butterfly territoriality (Davies, 1978) is sometimes interpreted outside of the biological literature as hinting at the presence of an endowment effect. Inside biology, that interpretation was refuted by subsequent experiments (Stutt & Willmer, 1998).

will address: how one could possibly think that biology is relevant to cognitive biases in the first place; why there are always two very different kinds of causes underlying behavior; and the basic connection between evolved behavioral predispositions and cognitive biases.

4.1 Biology is not just any discipline

Let's start with this obvious question: Why look to biology at all? One might be tempted to suppose, for instance, that the field of biology (with its subfields including evolutionary and behavioral biology) is just another field at an optional, lateral level of analysis—in the same way that economics, sociology, history, and political science are lateral and often fully autonomous fields within the social sciences. But that would be not just wrong, but rather couldn't-be-more-wrong wrong.

Here's why. The core principles of biology are as foundational to understanding the psychology and behavior of all organisms as the core principles of physics are foundational to understanding the movement of objects. To think otherwise is to imagine a fanciful world in which bodies—but not behavioral inclinations—evolve. A vast, immense, even staggering literature in biology—which I recognize may be unfamiliar to this journal's audience—makes that as absolutely clear as anything we know in any area of science (e.g., Futuyma & Kirkpatrick, 2023; Rubenstein & Alcock, 2018; Davies et al., 2012; Goldsmith & Zimmerman, 2001). There's more to it, of course. But in short, any body that arrives to this planet without behavioral predispositions is a dead end. That we live at all is a testament to the fact that behavioral predispositions can evolve and have evolved.

To flesh this out a bit more, let's start with the fact that human behavior comes from the brain. (Save only most reflexes, which are mediated through the spinal cord.) We know that the brain evolved, just as all the body's organs have evolved. And we know that the brain evolved to perform a variety of sophisticated functions. These include not only enabling top-down analytic decision-making, of the deliberative sort, but also bottom-up evolved shortcuts—which we colloquially call emotions—that quickly focus our attentions and predispose us toward behaviors that historically were both crucial and useful.

These behavioral predispositions are essentially information processing patterns that incline organisms who bear them toward behavioral tendencies when making choices in the face of uncertainty. Which is, after all, what all animals basically do all the time. Some basic biological perspectives are therefore necessary to ground an accurate and modern understanding of various kinds of behavioral leanings.

True, the behavior of living organisms is (obviously) less rigidly determined than is the behavior of inanimate objects, when forces act upon them. But that fact doesn't make living organisms magically immune to the underlying constraints, influences, and processes that enabled species, and the behavior of species, to exist in the first place.

Of course, the fact that principles of biology provide the foundation for the existence, psychologies, and behaviors of organisms doesn't mean that biology is somehow more important than, say, the field of psychology itself, to the goal of understanding behavior. Consider this analogy. When trying to understand how

planes work, it would be absurd to argue about whether physics or engineering is more important. Both are important. But just as engineering must ultimately be *compatible* with foundational principles of physics, psychology (and any other approach to decision-making and behavior) must also ultimately be compatible with foundational principles of biology.

For this reason, looking to biology for behavioral insights is not at all like looking to, say, sociology for behavioral insights. Biology is among the core natural sciences, alongside physics, chemistry, and a few others. Like all natural sciences, it is structurally, factually, and conceptually foundational in a way that social sciences, for all their differently added values, are not.

Am I suggesting that biology is more important than any other field? Not at all. I am saying, instead, that there are some puzzles about behavior that cannot be solved without understanding how evolutionary processes build brains and influence their behavior-generating patterns. Progress on puzzles such as these will often require teamwork, partnering the social sciences and life sciences together (Brosnan & Jones, 2023). Here's why.

4.2 There are always two kinds of causes

It has often been said (and endlessly quoted among biologists) that, “Nothing in biology makes sense except in the light of evolution” (Dobzhansky, 1973). Were there no evolution, there would be nothing alive. For our purposes, one of the most foundational principles of evolutionary biology—easily as fundamental as the relationship between supply and demand in economics—is one of which most social scientists are entirely unaware. It is a principle that eviscerates the fear that an evolutionary explanation for phenomena, such as cognitive biases, is somehow *in competition with* other favored explanations. And being unaware of this fundamental principle can quickly mire anyone interested in behavior in muddle-headed thinking.

The principle is this: When it comes to behavior, there are always two kinds of causes. I'll illustrate that in a moment, to make it vivid. But the key takeaway is that, in the ordinary course of analytic and scientific business, evolutionary explanations provide *complementary*, not *alternative*, explanations. Here's how that works.

Biologists draw a key distinction between *mechanistic* causes, which are termed *proximate causes* in biology, and *evolutionary* causes, which biologists call *ultimate causes* (Mayr, 1961; Vessey & Drickamer, 2019). The mechanistic causes are the physical pathways involving environmental conditions, neurons, hormones, chemical reactions, muscle movements, and the like. The evolutionary causes are the historical processes by which the mechanistic causes came to be more frequently elicited in some environmental circumstances than in others.

To see how these two kinds of causes are different but inevitably, unavoidably compatible, consider sex. Do people have sex because it feels good *or* because it is more likely to yield babies than many other activities in which one might indulge? The answer, of course, is: both. Both reasons are accurate at the same time.

It's true that you can't understand sexual pleasure without understanding how nerve endings are distributed in different arrangements in different parts of the body, how those nerve endings send impulses to the brain, and how the brain reacts to those

impulses. But it's also true that you cannot understand at any deeper level *why* those nerve endings are distributed as they are, and *why* those sensations tend to be considered pleasurable—rather than thoroughly revolting—without understanding the evolutionary relationship between sex and reproduction.

Put simply, those members of ancestral species in our lineage (hominid, primate, early mammalian, and so on) who found sex pleasurable tended, on average, to have more offspring (who inherited the tendency to find sex pleasurable) than those contemporaries who didn't. Then, by the inescapable mathematics of compounding, across hundreds of thousands of generations, finding sex pleasurable tends to be a feature shared very broadly across subsequent populations, who are almost entirely descended from individuals in the sex-feels-good group.

The important point to underscore is that it's *always* a profound error to pit *proximate* and *ultimate* explanations against each other. That's like assuming coins can be one-sided. Instead, one explanation of causes focuses on mechanisms, and the other focuses on why those mechanisms came to be as they are. These two kinds of causes are not only entirely complementary, they are in fact equally necessary. To think otherwise would be like pondering whether the area of a rectangle is the function of its length, on one hand, or of its width, on the other.³ Let's look at two examples.

First, what causes callouses? Recognizing that there are always two kinds of causes helps make plain that a callous is caused *both* by the skin encountering a persistent chafe (the mechanical cause) and by the fact that responding to chafing by thickening is an evolved adaptation of human skin (the evolutionary cause).

Second, why do some birds sing in the spring? A thorough answer requires attention to both kinds of causes. We know that lengthening days trigger hormonal changes that trigger forceful expulsion of air over particularly shaped vocal cords. That's the mechanistic cause. We also know that: (a) singing is attractive to mates (because, as it happens, singing well is a good proxy for strong, anti-parasite immune systems), and (b) spring is a more food-abundant time to raise offspring than are the depths of fall and winter. That's the evolutionary cause. So birds that sang, mated, and fed their offspring in the spring left more offspring than those who did those things in winter. And basically all the birds we see in the current population of that species are descended from those who sang in the spring, and inherited their behavioral predisposition to do so.

Getting used to thinking about behavior this way—that is, as always having two kinds of causes—helps clarify why widespread cognitive biases have evolutionary causes at the same time that they are sensitive to particular variables that are commonly-studied in behavioral economics and that are correlated with the appearance of behavior downstream from cognitive biases. For instance, a particular cognitive bias, such as the endowment effect, can have two kinds of causes simultaneously. It can be caused *both* by loss aversion *and* by how a propensity to loss aversion reflects our species' evolutionary history, running back even through shared ancestry with other primates for dozens of millions of years.

³ This metaphor is typically attributed to renowned psychologist Donald Hebb, during some Q & A following a public lecture. See, e.g., Meaney (2004).

Will I be suggesting that all human psychological biases are *adaptations*, that is, evolved features that became typical of species because they led demonstrably to gains in surviving, thriving, and reproducing? Not at all. Not every evolved feature—anatomical or psychological—is an adaptation shaped by natural selection. For instance, bones are white because calcium is white, not because white bones provide a reproductive advantage over purple bones. The same will be true of some psychological features. And there is a rich literature, stemming back to Williams (1966), on how to distinguish adaptations from non-adaptations. But some psychological features are best considered to have arisen as evolved adaptations. And there are reasons, as our experiments illustrate, to believe that at least some cognitive biases are evolved adaptations mis-matched to current environments.

4.3 Connecting evolution to cognitive biases

So how might evolutionary biology, resting largely on the way that natural selection shapes species, connect with cognitive biases? To see this, it's essential to first focus on something we already know and then focus on something we have long tended to overlook.

We already know that in many animal species, including humans (but excluding, say, jellyfish, which have a very simple nervous system), deciding on a course of action and then taking that action requires a brain. No shocker there. But we have tended to overlook several key facts about the relationship between brains, species-typical psychology, and behavior.

First, all brains, in all species, are the product of evolutionary processes that play out at the intersection of genes and environments. Second, the only evolutionary process that over time can yield so-called “adaptive” traits (which help organisms survive, thrive, and reproduce) is *natural selection*. I have so far referred to natural selection only in passing, assuming that most people are somewhat familiar with the concept from high school biology. But here's the core refresher, for current purposes. Natural selection unavoidably occurs wherever and whenever there is heredity (by inheritance), variation (by mutation and sexual recombination), and differential reproduction of offspring, as a function of heredity and variation. Third, natural selection not only shapes physical forms, *it also shapes some behavioral predispositions*. Think of those predispositions not as pre-determined, but rather as establishing inclinations, with strengths that vary (depending on the predisposition) from very weak to very strong. Fourth, natural selection is exquisitely adept at generating behavioral predispositions in brains by shaping every species' basic “psychology,” if you will, which is roughly typical of any given species, whether it is sharks, giraffes, or humans. Finally, natural selection can shape complex behavioral predispositions in extraordinarily subtle and sophisticated and algorithmic ways—the rich and vast detail of which is beyond my scope here. (For how all this works, see Futuyma & Kirkpatrick, 2023; Rubenstein & Alcock, 2018; Davies et al., 2012.)

It follows from these principles that, because natural selection is intimately involved in shaping some human behavior-biasing predispositions, at least some of the cognitive biases studied by behavioral economists might have evolved as adaptations. The key to understanding how cognitive biases that lead to irrational outcomes

could possibly be adaptations is this. It is quite common across evolutionary time for physical and behavioral traits that were once adaptive, given the environments in which they evolved, to become maladaptive in a later era, when there are different environmental conditions. This “mismatch” of a long-present adaptation to current environment is particularly likely when environments have changed very quickly on an evolutionary time-scale. For this reason, I and several others independently argued (in publications that were forthcoming at the same time) that many cognitive biases are likely to have originated in mismatched adaptations (Jones, 2001a; Haselton & Buss, 2000; Gigerenzer, 2000⁴).

Is it even possible to test a hypothesis about the evolutionary basis of a cognitive bias? It is possible. It has been done. The results are telling. And here’s how to understand both the approach and the findings.

5 Theoretical foundation part II: The evolution of the endowment effect

To my knowledge, I was the first to propose that the endowment effect is an evolved psychological adaptation to ancestral environments, now mismatched to rapidly changed current ones (Jones, 2001a, b). In order to frame this notion in relation to economists’ use of the term “rationality,” I found it useful to think of and discuss this mismatch as resulting in a special subset of irrationality—what I call a “time-shifted rationality”—by which a bias that is irrational today may have led to rational outcomes at an earlier time. But, whatever it’s called, the key notion is that the endowment effect may have evolved for the substantive advantages it provided in dramatically different environments than we now face.

For context, keep in mind that we humans have been primates for roughly 70 million years, have been our own species for merely 350,000 years, and adopted the modern calendar little more than 2,000 years ago. For all but the most recent blip of our primate history, it was inherently risky for a member of a social species like ours to give up an item in hand for the chance to trade it for something that might be slightly more desirable. Hence the aphorism: a bird in the hand is worth two in the bush (or, in this case, worth two in the hands of a potential trading partner).

Natural selection would therefore favor any psychological predisposition to discount the value of an item in another’s possession that might be acquired through trade compared to the value of what one already possessed. As stated earlier, but again here for emphasis, this attention to the ability of natural selection to generate physical and behavioral adaptations doesn’t mean that every physical and behavioral feature of an organism is an adaptation. Nor does it entail that the reproductive results be a conscious motive.

⁴ Ensuing decades have seen significant progress in exploring the adaptive roots of cognitive biases. See, for instance, Daly and Wilson (2005); Gigerenzer (2006); Haselton et al. (2009); Jackson & Willey (2011). The ABC Research Group at the Max Planck Institute for Human Development (directed by Gerd Gigerenzer) has been particularly prolific in this regard. The earliest work I’m aware of, to apply mismatch logic to any individual bias, is Kacelnik (1997).

Think about it this way. For all but the most recent eye-blink in our long evolutionary history, giving up something in hand for an attempted trade could result in having neither item in the end. Our modern architectures for enforcing trades after a trading partner's defection—such as contracts, laws for enforcement, police, lawyers, courtrooms, judges, and even language to describe a cheat to others—didn't exist.

In the same way that our predisposition to find sweets pleasurable originally evolved to motivate us to seek out scarce high-energy foods, but can lead to irrational behavior and bad outcomes in a world suddenly packed with cheap, ubiquitous, and ultra-high-calorie foods, our once-adaptive predisposition toward self-protective endowment effects can lead to irrational behavior in a world in which the riskiness of trades has gone way down due to our extremely recent but now largely effective modern enforcement mechanisms.

Sarah Brosnan and I set out to test the hypothesis that some inclination toward an endowment effect was an evolved adaptation. We and a changing set of colleagues conducted a series of experiments with the four novel predictions listed earlier, to the details of which we turn next.

6 Four novel predictions

6.1 Prediction 1: The endowment effect will be present in some other species, particularly close primate relatives

If a psychological predisposition toward manifesting an endowment effect in the face of various environmental conditions did in fact evolve in our species, it is unlikely that humans are the only species to have that adaptation. Sarah and I consequently predicted that chimpanzees (which are, along with bonobos, our closest primate relatives) would also exhibit an endowment effect. And our experiment testing this prediction provided the first direct evidence that the effect exists in a non-human species.⁵ Here's how the experiment worked.

Modifying a protocol that experimental economist Knetsch (1989) had used in humans, we created three different test conditions. Each offered chimpanzees a choice. In one condition, chimpanzee subjects—presented with a choice between two items, A and B—would point to the one they wanted and immediately get it. In the other two conditions, our subjects would either receive item A, and immediately be offered the option to trade it for item B, or instead receive item B, and immediately be offered the option to trade it for item A. Under these latter two conditions, chimpanzees would choose whether to keep what they had or to trade it for something else.

Using this paradigm enabled us to test for the presence of endowment effects without needing to quantify value in currency, as most human experiments do. If a given chimpanzee subject preferred item A to item B in the choice condition, and also separately received item A with the option to trade for B, we would of course expect our subject to refuse the offered trade. However, if that same subject who preferred item A to item B were given B, and the option to trade it for A, we would see evidence of

⁵ See fn 1.

an endowment effect if the subject did *not* trade B for A. In that case, in other words, having B in hand made it somehow more valuable to the subject than when it wasn't in hand. Which is to say: its value would have jumped from being less than the value of A to being more than the value of A as a function of possession. That's the tell-tale sign of an endowment effect.

These three conditions were of course counter-balanced among subjects—who could encounter them in any order—to avoid a possible confound of condition sequence. Also, because our first experiment tested two predictions simultaneously (the second of which is the subject of Sect. 6.2 below) we ran each of the three conditions with two different categories of objects. One category used foods, in which the A and B items were a flavored popsicle and a tube of peanut butter, respectively. The other category used toys, in which the A and B items were a dog's chew toy and a dog's rope toy.

There were three key findings from testing chimpanzees this way (Brosnan et al., 2007; Jones & Brosnan, 2008). Preferences in the choice condition tended to be quite stable, within subjects, over time. When first given their more-preferred food, our subjects essentially (and unsurprisingly) always kept it, instead of trading it for their less-preferred food. Most importantly, however, we discovered that many subjects tended to keep their *less*-preferred food when given the option to trade it for their *more*-preferred food.

That provided evidence that chimpanzees share endowment effect tendencies with humans. Not long after, other researchers explored related phenomena in other primate species (Lakshminaryanan et al., 2008; Kanngiesser et al., 2011). And subsequent work using our same experimental protocol found evidence of endowment effects in orangutans (Flemming et al., 2012) and gorillas (Drayton et al., 2013), which are the closest relatives to humans, after chimpanzees and bonobos.

6.2 Prediction 2: Chimpanzees will exhibit the endowment effect more frequently when trading foods for foods, than when trading toys for toys

If the endowment effect is an evolved adaptation, and if (as we know) natural selection can narrowly tailor adaptations to environmental contexts, it follows that the prevalence of the endowment effect will tend to vary across categories of items. More specifically, items high in *evolutionary salience* (such as foods, which provide energy and nutrition) will be more likely to yield endowment effects than will items of low evolutionary salience (such as toys). Here, evolutionary salience is essentially a measure of how much the possession of the item contributes to survival and reproduction compared to possession of alternative items. That is, if the consequences of not having an endowment effect are greater for some objects than others, natural selection would likely have positively correlated the strength of the behavioral bias with the relative disastrousness of not having it. We therefore compared the prevalence of endowment effects when subjects could trade foods for foods with the prevalence of endowment effects when subjects could trade toys for toys.

As predicted, across chimpanzees, orangutans, and gorillas (Brosnan et al., 2007; Drayton et al., 2013; Flemming et al., 2012) the endowment effect was far more prevalent—in chimpanzees, for instance, 14 times as prevalent—when subjects traded

foods for foods, compared to when trading toys for toys (where the effect was rather minimal).

Three important notes. First, this has nothing to do with the fact that subjects preferred food to toys. Of course they do. Instead, the important comparison is the prevalence of the endowment effect for toy-for-toy trading compared to the prevalence of the effect for food-for-food trading. Second, notice that if one had tested primate subjects without operating from an evolutionary framework, and had just tested them with toy-for-toy trading, one would have concluded—entirely erroneously—that the species did not exhibit endowment effects. Third, it would be a mistake to conclude from our data (showing that foods elicited endowment effects and toys didn't) that the evidence for endowment effects in these species is “mixed” (Apicella et al., 2014). That would miss the point entirely. For you can't test whether items high in evolutionary salience will be more likely to elicit an endowment effect than do items of low evolutionary salience without running an experiment that uses both, to enable the crucial contrast. What we showed here is that the prevalence of the endowment effect varies in somewhat predictable ways, for reasons solidly grounded in an explanatory theory.

6.3 Prediction 3: In chimpanzees, the endowment effect for the very same item will be more prevalent when that item can be used in context than when it can't

If natural selection biases behaviors in ways sensitive to the varying risks of losing items in a failed trade with a defecting trading partner, it follows that the prevalence of the endowment effect might also vary as a function of the immediate utility, or lack of utility, of the items in question. We tested this prediction by using tradable tools, with which our chimpanzee subjects were familiar, that only had immediate value when they could be used to extract different foods from containers. One tool could operate as a sponge to soak up otherwise unobtainable fruit juice. The other item was a dipstick that could be used to obtain otherwise unobtainable sweetened oatmeal.

We again tested chimpanzees across the three conditions of: choice between the two tools, offering to trade a sponge for a dipstick, and offering to trade a dipstick for a sponge. However, we also did all of those three under three contextual variants: (1) the two foods that the tools could extract were absent, (2) those foods were both present, but out of reach, and (3) the foods were both present and within reach.

As predicted, context mattered. A lot. Tool-for-tool trading elicited no endowment effect at all—zero—when the foods were absent. The same was true—zero endowment effect—when the food was present but out of reach. Yet as soon as the food was within reach, subjects exhibited a robust endowment effect for the tools, at essentially the same levels that they'd exhibited in our prior experiments. That is, when (and only when) the tools could be immediately useful, subjects frequently refused to trade their less-preferred tool for their more-preferred tool, and instead hung on to their less-preferred tool (Brosnan et al., 2012). Put another way, the prevalence of the endowment effect is not driven by the nature of the object alone, but can also depend on whether the tradable item is useful in context, in this case to acquire food.

6.4 Prediction 4: In humans, the evolutionary salience score of an object will predict some of the variance in the magnitudes of endowment effects, across a large set of novel items

The purpose of this entire arc of research, of course, was ultimately to learn something new about the patterns of endowment effects in humans, which would thereby illustrate a valuable approach linking some cognitive biases together, conceptually, by way of their likely-similar evolutionary origins. From the beginning, we set ourselves the goal of trying to predict some of the remarkably large variance in the magnitude of the endowment effect in humans. This was ambitious for the simple reason that—despite decades of research and many hundreds of published studies—none of the variation in endowment effect magnitudes had ever been successfully predicted for a large set of novel items. Indeed, I’m unaware of any prior studies that even attempted this.

Together with colleagues Chris Jaeger and Daniel Levin, and adapting methods of psychologists Gretchen Chapman and Eric Johnson (1995), we developed an experiment that worked with two different sets of human subjects (Jaeger et al., 2020). For one set, we created and provided a 6-question scoring survey, asking subjects to rate on a scale from 1 to 9 how much each of 24 items described to them contributes to a person’s health, increases a person’s attractiveness to potential dates, satisfies a basic human need, and the like.

We selected items we anticipated would provide a range of evolutionary salience from high (such as an entitlement to the best available healthcare, for life, for free) to low (such as a half-pound of Styrofoam packing peanuts). Summing the average ratings across all those prompts yielded an Evolutionary Salience Score for each of the items. To guard against bias, it was important that the scores of items be determined this way, that is, by objective third-party raters, not by any of us researchers.

The second set of subjects encountered the classic test for endowment effects (Horowitz & McConnell, 2002; Morewedge & Giblin, 2015). We divided subjects randomly into buyers and sellers and calculated the magnitude of the endowment effect for each hypothetically bought or sold item from the ratio of the buyer-group’s average maximum buy price to the seller-group’s average minimum sell price.

Given that there had been no published studies successfully predicting any of the variation in the magnitudes of endowment effects across a large set of novel items, we would have been thrilled if the combined evolutionary salience score of an item (summed, as indicated above, across several variables) predicted even 15% of the variation in endowment effect magnitudes across items. Instead, evolutionary salience scores—determined by third-party raters—predicted fully 52% of the variance (Jaeger et al., 2020).

Of course, our array of items was chosen specifically for the purpose of creating a wide range of evolutionary salience scores, so best to test this approach. As a result, we should not expect it to predict such a large percentage of variance for *all* future items under study. Still, this arc of experiments, and this most recent experiment in particular, suggests this evolutionary perspective and approach can do a significant amount of work both in explaining the evolutionary cause of, and in predicting patterns in, endowment effect magnitudes.

That, in turn, provides the first meaningful prospects not only for researchers but also for policymakers to begin to estimate more systematically and more accurately the likelihood that one thing or another will generate endowment effects sufficiently large to complicate economic and other policies. Put simply, the higher an evolutionary salience score, the more likely that citizens will overvalue what they already have, compared to what they could get in exchange.⁶

7 Conclusion

As Daniel Kahneman and his colleagues discovered, and many path-breaking behavioral economists have further explored, cognitive biases, leading to irrational choices and behavior, can throw sand in the gears of progress. For that reason, it's important to investigate and learn where biases come from and what influences the patterns in which they manifest. That may help equip us to better anticipate, deal with, and perhaps counteract some of them. Put simply, behavioral economists will be better predictors if they understand cognitive biases, and their two different kinds of causes, more deeply.

One long-running assumption, evident from the very isolation of the largely separate literatures on cognitive biases, is that most of the biases are disconnected phenomena, inconvenient one-offs. Another assumption, more implicit than explicit, has been that these are almost entirely human phenomena, best understood by studying humans alone. From a biological perspective, both assumptions are deeply suspect. Cognitive biases provide a window into behavioral predispositions. And widespread patterns in behavioral predispositions often (though importantly not always) reflect psychological adaptations that are typical of a given species and honed by the ubiquitous evolutionary force of natural selection to be useful in successfully confronting long-standing environmental challenges. Whenever environments change quickly on an evolutionary time-scale, it can lead to mismatches such that a once-adaptive predisposition can yield substantively irrational outcomes in a very new and meaningfully different current environment.

Understanding that two kinds of causes—both mechanistic and evolutionary—underlie the behaviors of all organisms can help deepen understandings that have been rooted in just either one alone. Understanding how natural selection works can help to explain and predict some deeply important patterns in economic behavior generally. After all, natural selection is the most relentlessly economizing force in the history of life (Jones et al., 2011). And inclining organisms to behave in ways that are sensitive to costs and benefits, under conditions of scarcity, is one of the most basic things natural selection does.

As I've mentioned elsewhere (Jones & Goldsmith, 2005), there are large literatures about the influence of natural selection on the origins of various tastes and preferences generally, as well as more specifically on patterns in cooperation and

⁶ Is it possible to overcome this primitive, once-adaptive, now out-dated bias? Of course. Experienced day-traders of stocks are less likely than members of the general population to exhibit the effect (List, 2003).

competition (including status competitions), costly third-party punishment, costly signaling, responses to incentives, the importance of relative as distinct from absolute well-being, cheater detection, and bounded rationality, among others (see e.g., Buss, 2015, 2025; Pinker, 1997; Fehr & Gächter, 2002). And even the briefest glance through an evolutionary lens raises the hypothesis that many cognitive biases may have arisen in deep ancestral primates as psychological adaptations to conditions now past.

The arc of research described here tested that hypothesis in a series of non-human and human experiments, focusing on the endowment effect as a test case. I do not wish to exaggerate the significance of the experiments described here. More work should be done, and alternatives explored (Bruner et al., 2020). But the fact remains that by incorporating just a few broadly accessible insights from evolutionary biology we were able to make a successful string of connected, theoretically grounded, coherent, step-wise predictions, none of which were predicted by any other extant theory.

Our approach found evidence strongly consistent with each of those predictions. Namely, other species, specifically close primate relatives, also show endowment effects. The effects are more prevalent for evolutionarily salient items than for less evolutionarily salient items. The immediate usefulness of an object can turn on an endowment effect where there had been none. And variations in evolutionary salience can, at least in the experimental conditions reported in our 2020 publication, predict a strikingly large amount of the variation in endowment effect magnitudes. All of this cross-coherent and converging evidence points to the endowment effect being an evolved psychological predisposition in humans and close primate relatives that is often irrational in the new human environments we've created for ourselves.

Note that this perspective does not supply an alternative explanation to loss aversion. Instead, it provides the necessary accounts of where loss aversion comes from in humans and related species, why it is patterned in particular ways, and how we can go about predicting and finding new patterns. While future research may uncover an even better scientific theory of the endowment effect, any such theory must now, at a minimum, account for the specific patterns observed in our closest primate relatives, who also display the effect, as well as accounting for known human patterns, in order to satisfy the criterion of parsimony, by which competing scientific explanations are judged.

The success of the approach summarized here not only deepens our understanding of the endowment effect. It also suggests more broadly that many other cognitive biases may be studied in similar fashion and potentially linked by a common conceptual understructure and deep theoretical framework. Looking at cognitive biases through the lens of natural selection may not only generate new predictions about patterns in existing biases, as well as predictions about biases yet to be discovered, but may also deepen the ways we understand the origins of, and patterns in, economic behavior.

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Declarations

Competing interests The author declares no competing interests.

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